

CS1231S Assignment #1

AY2023/24 Semester 2

Deadline: Wednesday, 21 February 2024, 1:00pm

ANSWERS

IMPORTANT: Please read the instructions below.

This is a graded assignment worth 10% of your final grade. There are **six questions** (“question” 0 and six problem-solving questions) with a total score of 40 marks. Please work on it by yourself, not in a group or in collaboration with anybody. Anyone found committing plagiarism (submitting other’s work as your own), or sending your answers to others, or other forms of academic dishonesty will be penalised with a straight zero for the assignment, and possibly an F grade for the module. Please see SoC website “Preventing Plagiarism” <https://www.comp.nus.edu.sg/cug/plagiarism/>.

You are to submit your assignment to **Canvas > Assignments** before the deadline.

Your answers may be typed or handwritten. Make sure that it is legible (for example, don’t use very light pencil or ink if it is handwritten, or font size smaller than 11 if it is typed) or marks may be deducted.

You are to submit a **SINGLE pdf file**, where each page is A4 size. Do not submit files in other formats. If you submit multiple files, only the last submitted file will be graded.

Late submission will NOT be accepted. We will set the closing time of the submission folders to slightly later than 1pm to give you a few minutes of grace, but in your mind, you should treat **1pm** as the deadline. If you think you might be too busy on the day of the deadline, please submit earlier. Also, avoid submitting in the last minute; the system may get sluggish due to overload and you will miss the deadline.

Note the following as well:

- Name your pdf file with your **Student Number**. Your student number begins with ‘A’ (eg: A0234567X). (Do not mix up your student number with your NUSNET-id which begins with ‘e’.)
- At the top of the first page of your submission, write your **Name** and **Tutorial Group**.
- To keep the submitted file short, please submit your answers without including the questions.
- As this is an assignment given well ahead of time, we expect you to work on it early. You should submit **polished work**, not answers that are untidy or appear to have been done in a hurry, for example, with scribbling and cancellation all over the places. Marks may be deducted for untidy work.
- It is always good to be clear and leave no gap so that the marker does not need to make guesswork on your answer. When in doubt, the marker would usually not award the mark.
- Do not use any methods that have not been covered in class. When using a theorem or result that has appeared in class (lectures or tutorials), please quote the theorem number/name, the lecture and slide number, or the tutorial number and question number in that tutorial, failing which marks may be deducted. Remember to use numbering and give justification for important steps in your proof, or marks may be deducted.

To combine all pages into a single pdf document for submission, you may find the following scanning apps helpful if you intend to scan your handwritten answers:

* for Android: <https://fossbytes.com/best-android-scanner-apps/>

* for iphone:

<https://www.switchingtomac.com/tutorials/ios-tutorials/the-best-ios-scanner-apps-to-scan-documents-images/>

If you need any clarification about this assignment, please do NOT email us or post on telegram, but post on **QnA “Assignments”** topic so that everybody can read the answers to the queries.

Question 0. (Total: 2 marks)

Check that ...

- you have submitted a pdf file with your Student Number as the filename. [1 mark]
- you have written both your name and tutorial group number (eg: T02) at the top of the first page of your file. (If you miss either one you will not be given any mark.) [1 mark]

Question 1. Propositional logic (Total: 7 marks)

To prove part (a), show each step with justification using only Theorem 2.1.1 and the implication law (if necessary). To disprove, cite a counter-example and show that it is a correct counter-example.

For part (b), you may use appropriate rules of inference in addition to Theorem 2.1.1 and implication law.

Do not use truth table for both parts. Do not cite any other results that have not been shown in class.

(a) Prove or disprove: $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$ [3 marks]

(b) Aiken and Dueet want to invite three friends Xes, Yul, and Zam to their party but are given these constraints:

1. Xes comes only if Yul does not come.
2. Yul comes if Zam or Xes comes.
3. Xes comes if Zam does not come.

- (i) Express the above constraints in propositional logic. [1 mark]
- (ii) Who will come to the party and who will not? Show how you arrive at your conclusion by giving a logical argument. (Not by writing an essay!) [3 marks]

Answers

(a) True. Proof:

$$\begin{aligned}
 & (p \vee q) \rightarrow r \\
 \equiv & \sim(p \vee q) \vee r && \text{(implication law)} \\
 \equiv & (\sim p \wedge \sim q) \vee r && \text{(DeMorgan's law)} \\
 \equiv & r \vee (\sim p \wedge \sim q) && \text{(commutative law)} \leftarrow \text{check if this step is skipped} \\
 \equiv & (r \vee \sim p) \wedge (r \vee \sim q) && \text{(distributive law)} \\
 \equiv & (\sim p \vee r) \wedge (\sim q \vee r) && \text{(commutative law x 2)} \leftarrow \text{check if this step is skipped} \\
 \equiv & (p \rightarrow r) \wedge (q \rightarrow r) && \text{(implication law)}
 \end{aligned}$$

(b) (i) Let x be “Xes comes to the party”, y be “Yul comes to the party”, and z be “Zam comes to the party”. Then the given constraints can be represented as follows:

Premise 1: $x \rightarrow \sim y$

Premise 2: $(z \vee x) \rightarrow y$

Premise 3: $\sim z \rightarrow x$

(ii)

1. Premise 2 is equivalent to $(z \rightarrow y) \wedge (x \rightarrow y)$ (by part (a)), hence we have

1.1. Premise 2.1: $(z \rightarrow y)$

1.2. Premise 2.2: $(x \rightarrow y)$

2. From Premises 1 and 2.2:

$$\begin{aligned} (x \rightarrow \sim y) \wedge (x \rightarrow y) &\equiv (\sim x \vee \sim y) \wedge (\sim x \vee y) && \text{(implication law x2)} \\ &\equiv \sim x \vee (\sim y \wedge y) && \text{(distributive law)} \\ &\equiv \sim x \vee (y \wedge \sim y) && \text{(commutative law)} \\ &\equiv \sim x \vee \text{false} && \text{(negation law)} \\ &\equiv \sim x && \text{(identity law)} \end{aligned}$$

3. From line 2 and Premise 3:

$$(\sim x) \wedge (\sim z \rightarrow x) \rightarrow z \quad \text{(modus tollens)}$$

4. From line 3 and Premise 2.1:

$$z \wedge (z \rightarrow y) \rightarrow y \quad \text{(modus ponens)}$$

5. Therefore, Yul and Zam will come to the party, while Xes will not.

Comments from grader (Vladimir):

(a) Missing commutative law.

(b) Missing generalization law. One common solution looks like this:

(P1) $x \rightarrow \sim y$

(P2) $(z \vee x) \rightarrow y$

(P3) $\sim z \rightarrow x$

Prove by division by cases on x .

1. Case 1: Assume x holds.

1.1. $\sim y$ (by modus ponens on P1 and x)

1.2. y (by modus ponens on P2 and x) $\leftarrow$ generalization rule application is missing here!

1.3. Contradiction (from 1.1 and 1.2)

2. Case 2: Assume $\sim x$ hold.

2.1. $\sim x \rightarrow \sim(\sim z)$ (contrapositive of P3)

2.2. $\sim(\sim z)$ (by modus ponens on 2.1 and $\sim x$)

2.3. z (double negation law on 2.2)

2.4. $z \vee x$ (by generalization law on 2.3)

2.5. y (by modus ponens on P2 and 2.4)

Hence, only Zam and Yul will come to the party.

The mistake is at 1.2, because x doesn't exactly match the hypothesis of P2, which is $(z \vee x)$. Instead, one should apply generalisation rule first (as it is done at step 2.4).

Question 2. Argument (Total: 7 marks)

- (a) Determine the validity of the following argument: [2 marks]

“Zack does not have a keen sense of hearing. Zack is a pianist or a writer. Zack is a pianist only if he has a keen sense of hearing. Therefore, Zack is a writer.”

You may use the following statement variables:

p : Zack is a pianist.

w : Zack is a writer.

h : Zack has a keen sense of hearing.

- (b) Given a particular tour group, the following argument is presented:

“A tourist in this tour group has not visited Gardens by the Bay. Every tourist in this tour group has visited Sentosa. Therefore, someone in this tour group who has visited Sentosa has not visited Gardens by the Bay.”

Define appropriate predicates and determine the validity of the above argument. (Write a proper proof, not an essay!) Let the domain of discourse be the set of all human beings and you may omit this set in your statements. [5 marks]

Answers

- (a) Note that “or” in English means the “inclusive or” in logic. The argument is valid. Proof:

1. $\sim h$ (premise 1)
2. $p \vee w$ (premise 2)
3. $p \rightarrow h$ (premise 3)
4. $(p \rightarrow h) \wedge \sim h \rightarrow \sim p$ (modus tollens on 3 and 1)
5. $(p \vee w) \wedge \sim p \rightarrow w$ (elimination on 2 and 4)
6. Hence the argument is valid.

Should have included the instruction “Do not use truth table/critical rows method for this problem”.

- (b) The argument is valid. Proof:

1. Let the universal set U be the domain. Define predicates:
 - 1.1. $T(x) \equiv x$ is a tourist in the tour group.
 - 1.2. $G(x) \equiv x$ visited Gardens by the Bay.
 - 1.3. $S(x) \equiv x$ visited Sentosa.
2. $\exists x \in U (T(x) \wedge \sim G(x))$ (premise 1)
3. $\forall x \in U (T(x) \rightarrow S(x))$ (premise 2)
4. $T(a) \wedge \sim G(a)$ for some $a \in U$ (by existential instantiation on 2)
5. $T(a)$ for some $a \in U$ (by specialization on 4)
6. $S(a)$ for some $a \in U$ (by universal modus ponens on 3 and 5)
7. $T(a) \wedge \sim G(a) \wedge S(a)$ for some $a \in U$ (by conjunction on 4 and 6)
8. $T(a) \wedge S(a) \wedge \sim G(a)$ for some $a \in U$ (by commutative law) ← to grader: may be skipped
9. $\exists x \in U (T(x) \wedge S(x) \wedge \sim G(x))$ (by existential generalisation on 8)

The universal set may be assumed and omitted. Hence, the above may be written as:

1. Define predicates:
 - 1.1. $T(x) \equiv x$ is a tourist in the tour group.
 - 1.2. $G(x) \equiv x$ visited Gardens by the Bay.
 - 1.3. $S(x) \equiv x$ visited Sentosa.
2. $\exists x (T(x) \wedge \sim G(x))$ (premise 1)
3. $\forall x (T(x) \rightarrow S(x))$ (premise 2)
4. $T(a) \wedge \sim G(a)$ for some a (by existential instantiation on 2)
5. $T(a)$ for some a (by specialization on 4)
6. $S(a)$ for some a (by universal modus ponens on 3 and 5)
7. $T(a) \wedge \sim G(a) \wedge S(a)$ for some a (by conjunction on 4 and 6)
8. $T(a) \wedge S(a) \wedge \sim G(a)$ for some a (by commutative law) ← to grader: may be skipped
9. $\exists x (T(x) \wedge S(x) \wedge \sim G(x))$ (by existential generalisation on 8)

The question suggests letting the domain of discourse be all humans. But some students defined their own domain, which I accepted. (My intention is to let students work with universal conditional statement, and apply universal modus ponens.)

So, alternatively, we may define the domain of discourse to be T , the set of tourists. Then the proof will look like this.

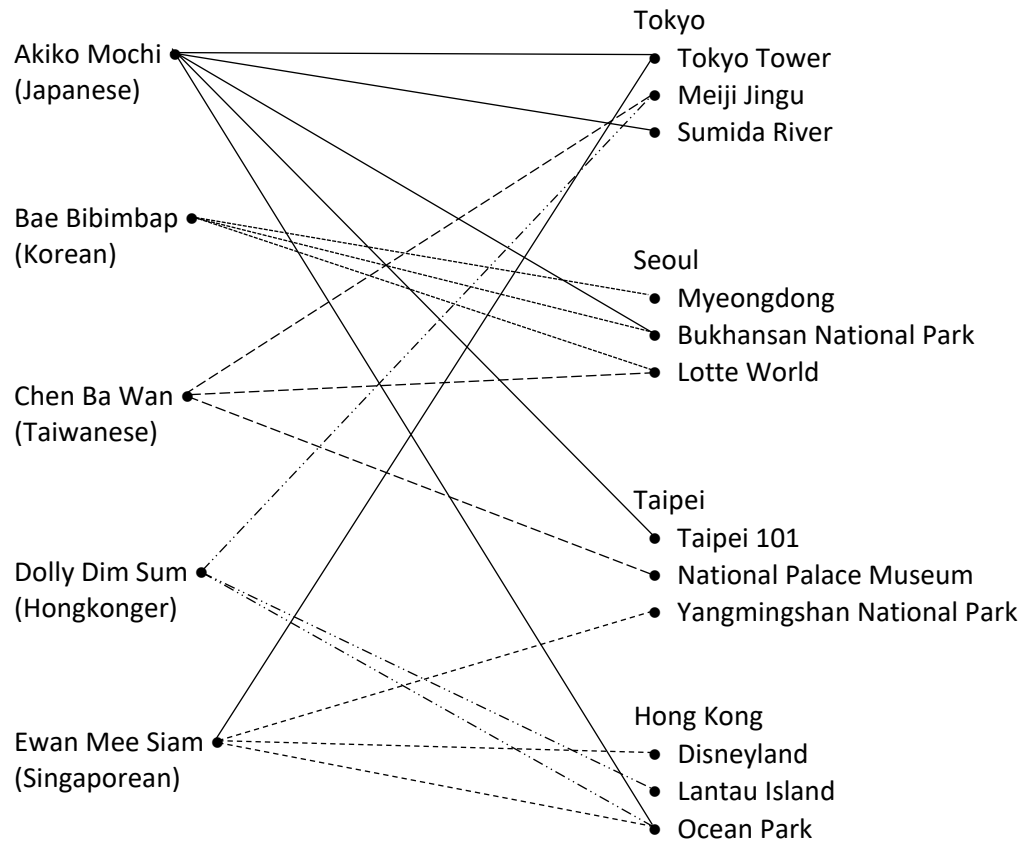
1. Let T be the set of tourists. Define predicates:
 - 1.1. $G(x) \equiv x$ visited Gardens by the Bay.
 - 1.2. $S(x) \equiv x$ visited Sentosa.
2. $\exists x \in T \sim G(x)$ (premise 1)
3. $\forall x \in T S(x)$ (premise 2)
4. $\sim G(a)$ for $x = a$ (by existential instantiation on 2)
5. $S(a)$ for $a \in T$ (by universal instantiation on 3)
6. $S(a) \wedge \sim G(a)$ for $a \in T$ (by conjunction on 5 and 4)
7. $\exists x \in T (S(x) \wedge \sim G(x))$ (by existential generalisation on 6)

Comments from grader (Aaron):

- A handful of students solved part (a) using the critical rows method, which is not my intention. However, as I didn't have an instruction forbidding it, I accepted the answer. The critical rows method does not showcase students' mastery of the rules of inference.
- Part (b) was poorly attempted. I have put comments on individual submissions to highlight the mistakes, so please go over them.
- Writing an argument is a skill that required constant practice. Your first draft is unlikely a good proof, so go over your proof a number of times, revising and polishing it.
- Despite the instruction ("Write a proper proof, not an essay!"), there were a number of students who wrote the proof mostly in English without any/much mathematical notations.
- A number of students provided no or very few justifications in their proof.
- A number of students misquoted some rule, for example, citing "modus ponens" where it should be "universal modus ponens".
- Some students used proof by contradiction. When a simple direct proof exists, use it. Use indirect proof when a direct proof is impossible or hard.

Question 3. Quantified statements (Total: 8 marks)

- (a) The diagram below shows the places of interest in four cities where five visitors went to. You may assume that the Japanese is resident in Tokyo, the Korean in Seoul, and the Taiwanese in Taipei. In the context of this question, the places of interest listed are all the places of interest in that city. A line joining a visitor to a place of interest indicates that the visitor visited that place.



Answer the following parts with “true” or “false”, and give your explanation clearly. [5 marks]

- There is some visitor who visited all the places of interest in some city.
- Every visitor visited at least two places of interest in some city.
- Some of the places of interest in every city were visited solely by non-residents of that city.
- Every visitor visited some places of interest that were also visited by another visitor.
- The place of interest visited by the most visitors is located in the city visited by the most visitors.

- (b) Given the following predicates:

- $Cat(x)$: x is a cat.
- $Dog(x)$: x is a dog.
- $Socialist(x)$: x is a socialist.
- $Likes(x, y)$: x likes y .

Write a quantified statement for each of the following sentences. You may assume that the domain of discourse is the set of all animals A . [3 marks]

- No cat is a socialist.
- All cats are liked by some dog.
- No dog likes a socialist.

Answers

(a)

- (i) True. BaeBibimbap visited all the places of interest in Seoul.
- (ii) False. Chen Ba Wan visited at most one place of interest in every city.
- (iii) False. None of Seoul's places of interest were visited solely by non-Koreans.
- (iv) True. Akiko Mochi visited Tokyo Tower visited by Ewan Mee Siam; Bae Bibimbap visited Lotte World visited by Chen Ba Wan; Chen Ba Wan visited Lotte World visited by Bae Bibimbap; Dolly Dim Sum visited Ocean Park visited by Ewan Mee Siam; Ewan Mee Siam visited Tokyo Tower visited by Akiko Mochi.
- (v) False. The most visited place of interest is Ocean Park. However, the most visited city is Tokyo.

(b)

- (i) $\forall x \in A (Cat(x) \rightarrow \sim Socialist(x))$.
- (ii) $\forall x \in A (Cat(x) \rightarrow \exists y \in A (Dog(y) \wedge Likes(y, x)))$.
- (iii) $\forall x, y \in A ((Dog(x) \wedge Socialist(y)) \rightarrow \sim Like(x, y))$ or
 $\forall x \in A (Dog(x) \rightarrow \forall y \in A (Socialist(y) \rightarrow \sim Like(x, y)))$.

Comments from grader (Justin):

- (a) (i) Almost all got it right.
- (ii) Many failed to address that Chen Ba Wan visited at most one place of interest in **every** city. Some listed out the number of places Chen Ba Wan visited in each city but left out that Hong Kong was not visited. Since every city should be accounted for, marks were deducted if such.
- (iii) Almost all got it right.
- (iv) Some did not give an explanation, and just reiterated the statement.
- (v) Some incorrect explanations, incorrect most visited city. Some counted the number of visits instead of the number of visitors.
- (b) (i) Almost all got it right.
- (ii) Missing domain and brackets were common for this, especially for the existential quantifier.
- (iii) Some were unable to get the final answer. Wrongly answered "conjunction" instead of "implication law". Surprisingly some misconception in writing quantified statements -> using "nested functions": Likes(dog(x), socialist(y)) which is wrong. [Aaron's comment: This is highlighted in "Tutorial 2 Common Mistakes"!]
- Some left out parentheses for the conclusion, thus making the statement logically ambiguous.

Question 4. (Quantified statements on sets) (5 marks)

Let $A = \{-1, 0, 1\}$, $B = \{0, 1, 4\}$ and $C = \{-2, -1, 0, 1, 2\}$. Indicate whether each of the following statements is true or false with explanation.

- (a) $\exists x \in A \forall y \in B \exists z \in C ((y \neq 0) \wedge (y > xz))$.
- (b) $\exists x \in A \forall y \in B \exists z \in C (z > xy)$.
- (c) $\forall x \in C ((\forall y \in B xy \in B) \rightarrow (x^2 = x))$.

Answers:

(a) False. Since $0 \in B$, it is not possible to have $y \neq 0 \forall y \in B$.

(b) True.

The required x is -1 or 0 . (Only one is needed.)

For $x = -1$, for all $y \in B$, there exists $z = 1$ (or $z = 2$) such that $z > -y$.

For $x = 0$, for all $y \in B$, there exists $z = 1$ (or $z = 2$) such that $z > 0$.

(c) True.

Case $x = -2, -1$ or 2 : $(\forall y \in B xy \in B)$ is false (counterexample: $y = 4$),

therefore $((\forall y \in B xy \in B) \rightarrow (x^2 = x))$ is vacuously true.

Case $x = 0$ or 1 : $(\forall y \in B xy \in B)$ is true and $(x^2 = x)$ is also true,

therefore $((\forall y \in B xy \in B) \rightarrow (x^2 = x))$ is true.

In all cases, $((\forall y \in B xy \in B) \rightarrow (x^2 = x))$ is true

Marking scheme: 2 marks. True (1 mark), justification (1 mark: ½ mark per set of cases).

Students may list out all 5 cases individually. Award justification marks only if students got the truth value correct.

Comments from grader (Shanmugam Udhaya):

(a) Mostly well done.

(b) Badly done.

Many claimed it is true but the method to get the result is not completely correct. Here the given statement is saying there exists an x such that for all y , you can find a z where $z > xy$. So what we are expecting is for students to pick an x value first, then consider all cases of y , then finally pick an z .

Instead, many picked an x and z from the start due to the intuition that picking that z will no matter what result in $z > xy$. While they lead to the same conclusion, it is not strictly following the order of the quantifiers. The student's method would be for $\exists x \in A \exists z \in C \forall y \in B$ instead.

[Aaron's comment: It is important to stick to the order of the quantifiers. This is a common mistake that many students are unaware of, so this is a good reminder.]

(c) Badly done as well due to misinterpretation of the conditional statement.

To correctly interpret the conditional statement, it is for all x in C , IF considering all y , xy is found in B , then we check whether $x^2 = x$. The "for all y " is important, suppose you found out that for some x and for some y , xy such as $xy = 8$, is not in B , the hypothesis becomes false and the conditional statement as a whole becomes (vacuously) true!

What students have done mostly is pick a counter-example to show for all x , there is some xy in B then show $x^2 \neq x$. For example, when $x = 2$, $y = 0$, $xy = 0$ is found in B then show $4 \neq 2$. This is wrong as the conditional statement has to check for every y and not just ONE y .

So the right way is to consider for all x , breaking down into the cases of x in $\{1,0\}$ and x in $\{-2, -1, 2\}$, as shown in the answer. Some caught on to the idea but only considered one of the two cases.

Question 5. Sets (5 marks)

Prove or disprove the statements below. Do not use Venn diagrams in your proof or counter-example.

(a) $\overline{A \setminus (B \cup C)} \subseteq \bar{A} \cap (B \cup C)$ for all sets $A, B, C \subseteq \mathbb{Z}$.

(b) $(A \cup B) \setminus (C \setminus A) = A \cup (B \setminus C)$ for all sets A, B, C .

Answers:

(a) False. 1 mark.

Counter-example: Let $A = B = C = \{1\}$, then $\overline{A \setminus (B \cup C)} = \mathbb{Z} \not\subseteq \emptyset = \bar{A} \cap (B \cup C)$.

(b) True. 4 marks.

1. $(A \cup B) \setminus (C \setminus A) = (A \cup B) \cap (\overline{C \setminus A})$ (set difference law)
2. $= (A \cup B) \cap (\overline{C \cap \bar{A}})$ (set difference law)
3. $= (A \cup B) \cap (\bar{C} \cup \bar{\bar{A}})$ (De Morgan's law)
4. $= (A \cup B) \cap (\bar{C} \cup A)$ (double complement law)
5. $= (A \cup B) \cap (A \cup \bar{C})$ (by commutative law)
6. $= A \cup (B \cap \bar{C})$ (by distributive law)
7. $= A \cup (B \setminus C)$ (by set difference law)

Comments from grader (Chin Herng):

(a) A number of students came up with a hypothetical element x that belongs to certain sets and does not belong to certain other sets, then used this element as a counter-example to disprove the subset relationship. This approach mostly works, but most students did not consider the case where such an element x does not exist. For example, students would write "let $x \in A$ " without considering the case where A is empty.

Some students also realised that if the given subset relationship were to be reversed, then the universal statement would be true. This is not an immediate negation of the given statement to disprove. In particular, without further work, it remains a possibility that both containments ($\subseteq$ and $\supseteq$) are true, resulting in set equality.

(b) Commonly skipped justifications are the Double Complement Law and the Commutative Law.

A number of students did not use equal signs correctly between each line. Some incorrectly used $\equiv$ or $\leftrightarrow$ which are mostly relevant for logical statements only. The expected answer is a chain of equalities from the LHS to the RHS. Some students simplified LHS and RHS separately to the same expression. This is also accepted.

Question 6. Writing proof (6 marks)

Prove the following statement:

“If the square of an integer is divisible by 5, then that integer is divisible by 5.”

(Do not write proof that is not rigorous. For example, “if n^2 is divisible by 5, then n^2 ends with 0 or 5 (everybody knows this), and so n must also end with 0 or 5 (common sense).” Also do not use real number functions such as square root. Use only what have been covered in class.)

Answer

Proof by contraposition.

1. The contrapositive of the given statement is: $\forall n \in \mathbb{Z} (5 \nmid n \rightarrow 5 \nmid n^2)$.
2. Let x be an integer that is not divisible by 5.
3. By Theorem 4.4.1 Quotient-Remainder Theorem, x can be written in one of these 5 forms (Lecture 5 slide 32): $x = 5k$ or $x = 5k + 1$ or $x = 5k + 2$ or $x = 5k + 3$ or $x = 5k + 4$ for some integer k . Except for the first case where x is divisible by 5, the rest are not.
 - 3.1. Case $x = 5k + 1$: Then $x^2 = (5k + 1)^2 = 25k^2 + 10k + 1 = 5(5k^2 + 2k) + 1$.
Note that $5k^2 + 2k \in \mathbb{Z}$ by closure of integers under $\times$ and $+$.
 - 3.2. Case $x = 5k + 2$: Then $x^2 = (5k + 2)^2 = 25k^2 + 20k + 4 = 5(5k^2 + 4k) + 4$.
Note that $5k^2 + 4k \in \mathbb{Z}$ by closure of integers under $\times$ and $+$.
 - 3.3. Case $x = 5k + 3$: Then $x^2 = (5k + 3)^2 = 25k^2 + 30k + 9 = 5(5k^2 + 6k + 1) + 4$.
Note that $5k^2 + 6k + 1 \in \mathbb{Z}$ by closure of integers under $\times$ and $+$.
 - 3.4. Case $x = 5k + 4$: Then $x^2 = (5k + 4)^2 = 25k^2 + 40k + 16 = 5(5k^2 + 8k + 3) + 1$.
Note that $5k^2 + 8k + 3 \in \mathbb{Z}$ by closure of integers under $\times$ and $+$.
4. In all cases, x^2 is not divisible by 5.
5. Therefore, the statement is true.

Comments from grader (Eom Ikhoon):

- Directly concluding without reducing r^2 into values between 0 and 5 that n is not divisible by 5; you need to reduce r^2 into values between 0 and 5 to conclude (by existence and uniqueness of quotient and remainder by Theorem 4.4.1) that indeed n^2 is not divisible by 5.
- Abuse of notation. Many students let $q = 5p^2 + 2pr$ and used q for the entire proof. Note that the choice of q is not uniform – q depends on the value of r .
- No mention of the quotient-remainder theorem. This theorem is the key theorem that drives the proof. Without proper citation of this theorem, your proof is not validated since the proof implicitly assumes that quotient and remainder are unique for the whole procedure, and they exist, which can be partitioned into the 4 cases of $r = 1, 2, 3, 4$. Many students mentioned the definition of divisibility, but this is not the same as the quotient-remainder theorem.
- Implicitly assuming the Fundamental Theorem of Arithmetic, which is not covered in class. Students must prove the Fundamental Theorem of Arithmetic in order to use it. There are proofs submitted by students that implicitly apply this theorem.

=== End of paper ===